

Learning Activity Sheet for General Mathematics

Quarter 1

LAS

3

Learning Activity Sheet for Mathematics Grade 8
Quarter 1: Unit 2

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Development Team

Writer: Kwenie A. Paje
Region II

Validator: Emily Tabamo
Region X

Language Editor:

Consultant: Dr. Flordeliza F. Francisco
Far Eastern University

Learning Area Specialist: Wilson R. Santiago

Bureau of Learning Delivery
Bureau of Curriculum Development
Bureau of Learning Resources

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph.

LEARNING ACTIVITY SHEET

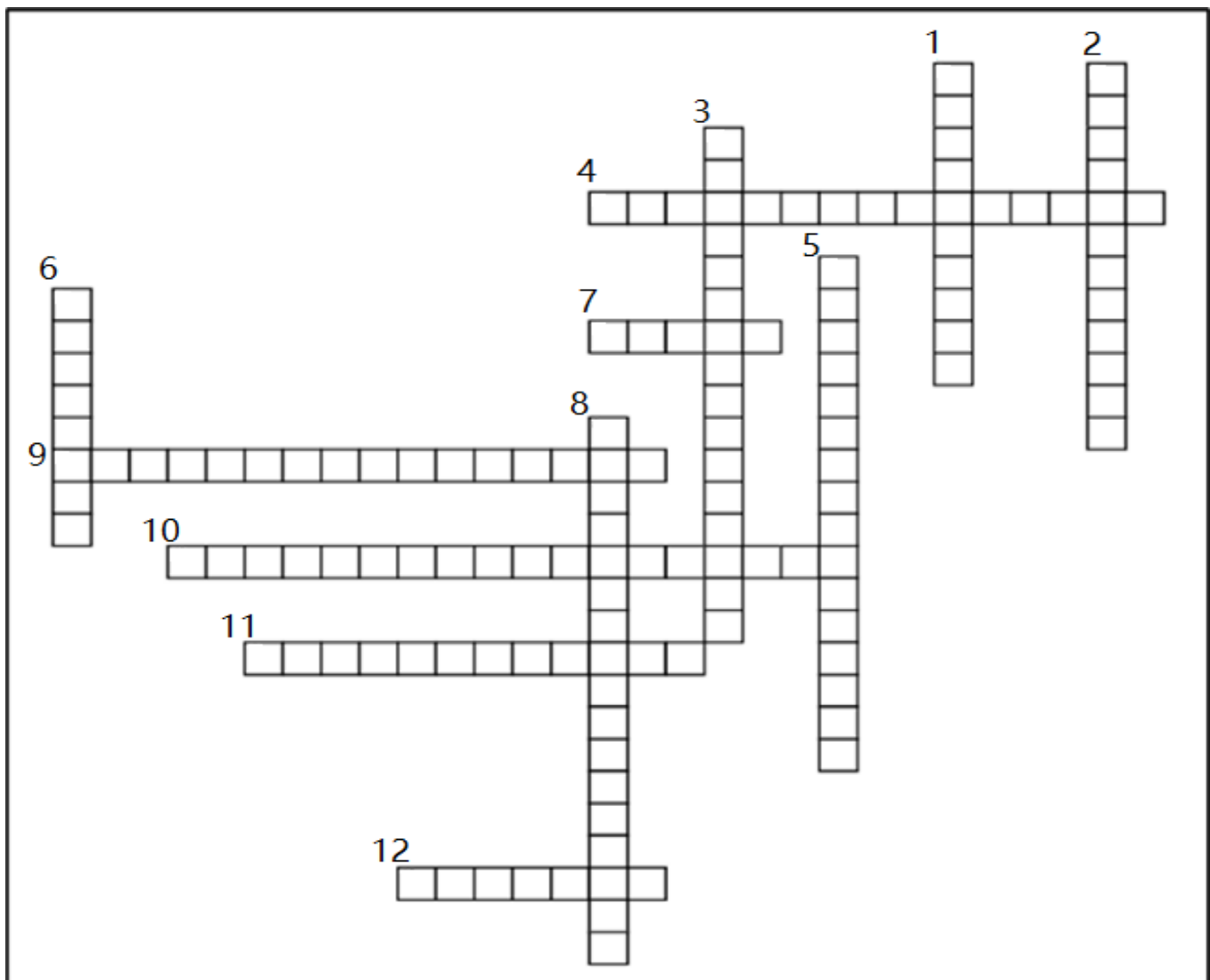
Learning Area:	General Mathematics	Quarter:	1
Lesson No.:	3	Date:	
Lesson Title/ Topic:	Applications of Sequences and Series in Financial Mathematics: Understanding Appreciation, Depreciation, Compound Interest, Loans, Mortgage Payment and Investment Growth		
Name:		Grade & Section:	

I. Activity: Crossword Puzzle

- Objective(s):** At the end of the activity, you should be able to identify real-world financial applications involving sequences and series

II. Materials Needed: pen and paper

III. Instructions: Answer the crossword puzzle by starting with the easiest clues to uncover letters that will help solve the more difficult ones



ACROSS

- 4. Interest earned on past interest..
- 7. Borrowed money with interest.
- 9. Sum of an arithmetic sequence interest.
- 10 Add same number repeatedly.
- 11 Increase in asset value.
- 12 Money paid regularly for debt.

DOWN

- 1. Money saved to earn profit.
- 2. Decrease in asset value.
- 3. Interest earned on past
- 5. Principal plus earned interest.
- 6. Loan for buying property.
- 8. Multiply by constant ratio.

IV. Synthesis:

1. What strategies did you use to solve the crossword puzzle, and how did they help you arrive at the correct answers?

2. How did you feel while doing the activity? What aspects of solving the crossword puzzle did you enjoy or find challenging?

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I. Activity : Problem – solving on Financial Problems Using Series/Sequence

Objective(s): At the end of the activity, you should be able to:

1. Determine the possible types of sequences and series that could be applied in financial applications.
2. Solve real-life financial problems by applying sequences and series

II. Materials Needed: pen and paper

Instructions: Solve the following problems step by step. Make sure to follow the correct process.

1. A farmer borrowed ₱30,000 at 8% interest compounded annually. How much will he have to pay at the end of 5 years?

1: Identify what is being asked
2: Identify the variables
STEP 3: Identify the Given values
STEP 4: Identify the series / sequence to be used
STEP 5: Show the solution
STEP 6. Provide the answer

2. A company's annual profit at the end of 2022 was ₱210,000. The profit is expected to increase by 4% each year. Calculate the company's expected annual profit by the end of 2027 and the accumulated profit from 2022-2027.

1: Identify what is being asked
2: Identify the variables
STEP 3: Identify the Given values

STEP 4: Identify the series / sequence to be used
STEP 5: Show the solution
STEP 6: Provide the answer

3. Liam starts a scholarship fund to support students from rural areas. As the first donor, he donates ₱200,000. Each new donor contributes ₱50,000 more than the donor before them. How much does the 12th donor contribute to the fund? What is the total amount donated by the first 12 donors?

1: Identify what is being asked
2: Identify the variables
STEP 3: Identify the Given values
STEP 4: Identify the series / sequence to be used
STEP 5: Show the solution
STEP 6: Provide the answer

4. A community fundraiser begins with an initial donation of ₱15,000. Each supporter contributes a fixed amount of ₱3,000 to the fund. When the 40th person makes their contribution, how much money is now in the fund?

1: Identify what is being asked
2: Identify the variables
STEP 3: Identify the Given values

STEP 4: Identify the series / sequence to be used
STEP 5: Show the solution
STEP 6: Provide the answer

5. If the Principal ₱2,000 and the interest rate is compounded 8% annually. Calculate the account balance after 4 years.

1: Identify what is being asked
2: Identify the variables
STEP 3: Identify the Given values
STEP 4: Identify the series / sequence to be used
STEP 5: Show the solution
STEP 6: Provide the answer

V. Synthesis:

- What strategies did you apply to solve the financial problems, and how were these strategies similar or different from those you used in the crossword puzzle? How did these approaches help you understand the concepts better?

- How did the activity affect your interest and confidence in learning financial mathematics? What parts did you find most enjoyable or challenging, and why?
